

Healthcare Transparency MVP

Master Plan v3 — NYC Orthopedic Value Index

Direct-to-Consumer Healthcare Transparency Marketplace | Revision 3.0 | August 2026

1. Executive Summary

This document supersedes Master Plan v1 and v2. The strategic thesis is unchanged: consumer-directed healthcare has created real demand for a platform that makes clinical outcomes and financial liability legible in the same view, and New York's public clinical data combined with federally mandated pricing files is enough to build one.

What has changed is that the pipeline has now been **executed against live data**, and that execution invalidated a premise carried through every previous version of this plan.

The procedure codes in the original plan do not exist. Every prior version scoped the MVP to APR-DRG 301 (hip) and 302 (knee). Queried against both the 2022 and 2024 SPARCS releases, those codes return **zero rows**. The correct codes are **APR-DRG 324 (elective hip)** and **326 (elective knee)**. Had this gone undetected, the pipeline would have run cleanly end to end and produced an empty market.

Forty defects have now been identified and corrected across the SQL and Python: eight of them found only by running the code against the real Health Data NY API and real hospital price files. Section 3 records them, because a portfolio project that cannot audit its own pipeline is not demonstrating the skill it claims to.

Scope

- **Procedure** — elective total hip and knee replacement. **APR-DRG 324 and 326** in the state clinical data; **MS-DRG 470**, or APR-DRG 324/326, in the federal pricing files.
- **Geography** — the five boroughs of New York City, scored against a statewide clinical benchmark.
- **Output** — a risk-adjusted Value Index per facility, published as an interactive web application and a geographic dashboard.

2. Current Status

Honest accounting of what exists as of this revision.

Component	Status	Notes
Python ETL	Verified against live data	All modules import and compile. SPARCS API and CMS crawler both exercised end to end.
CMS MRF crawler	Working	14 of 18 target domains resolved; 70+ machine-readable file locations discovered.
MRF parsers	Working	JSON and CSV (tall and wide) all parse real files and return real prices.
SQL pipeline	Written, awaiting DB	Six idempotent scripts plus a self-asserting smoke test. Blocked only on database credentials.
Public web app	Built, not deployed	Streamlit. Reads a cloud database when present, a committed CSV snapshot otherwise.
Facility crosswalk	Automated, needs review	Exact and fuzzy matching implemented; the fuzzy tier requires human sign-off.
GitHub repository	Published	Public. Credential scan clean: no secrets tracked, none in history.
Tableau dashboard	Not started	Depends on a completed pipeline run.
Figma wireframes	Not started	Independent of the data work; can proceed in parallel.

The critical path now runs through a single item: the PostgreSQL password, without which no SQL stage can execute.

3. Corrections to the Original Plan

Forty defects were identified. The full catalogue lives in the repository at `docs/ERRATA.md`. The ones that changed the design, rather than merely fixing a line of code, are below.

3.1 The procedure codes were wrong

APR-DRG 301 and 302 return nothing in SPARCS; 303 is a lumbar fusion. The APR-DRG v38 classification the data actually uses is:

Code	Description	In scope?
323	NON-ELECTIVE OR COMPLEX HIP JOINT REPLACEMENT	No
324	ELECTIVE HIP JOINT REPLACEMENT	Yes
325	NON-ELECTIVE OR COMPLEX KNEE JOINT REPLACEMENT	No
326	ELECTIVE KNEE JOINT REPLACEMENT	Yes

This is a genuine improvement on the original design as well as a correction: 324 and 326 carry the elective distinction in the code itself, so the cohort is defined by the classification system rather than inferred from an admission-type filter.

3.2 APR-DRG 470 is not MS-DRG 470

NYU Langone publishes every DRG line in its price file as APR470-1 through APR470-4 — the suffixes are APR severity subclasses — with a declared code type of merely DRG. Any matcher looking for "470" reads that as MS-DRG 470 and imports an entirely unrelated procedure's price into the joint-replacement median.

Billing codes are now resolved to a *(system, number)* pair. The two DRG systems are never pooled, and APR-DRG 324/326 is accepted as a legitimate alternative source of pricing, because it maps to exactly the cohort the clinical side already uses.

3.3 Facility identity is not the facility name

Every facility name changed case between the 2022 and 2024 releases, and some changed form entirely — New York - Presbyterian/Queens became NEWYORK-PRESBYTERIAN/QUEENS. Grouping on the name splits one hospital into two half-volume rows, computing both O/E ratios against the wrong denominators and halving the experience modifier.

The Permanent Facility Identifier was verified 100 percent populated in both releases, with all 45 of the 2022 facilities present in 2024. Facility metrics, the crosswalk and the master join are now keyed on the

PFI, with the display name taken from the most recent year. This is what the original plan recommended; the original code never retrieved the column.

3.4 The borough is spelled differently each year

hospital_county is Manhattan in the 2022 release and New York in 2024. A single-spelling county list silently drops an entire borough from one year. The filter now accepts both, and additionally matches the service-area column — which is New York City in both years, but is named hospital_service_area in 2022 and health_service_area in 2024.

3.5 Defects that would have produced confident, wrong numbers

Defect	Consequence had it shipped
Disposition matched on the substring "Home"	Skilled nursing and hospice discharges scored as clinical successes.
Pricing joined at the wrong grain	Facility rows multiplied per billing code, corrupting the market median.
Billing code matched as a substring	Unrelated codes, including a liver procedure, entered the price median.
Negotiated column selected by loose match	A negotiated <i>percentage</i> written into a dollar column as a \$65 knee replacement.
Pandas index misalignment	Description, payer and both price columns silently null after the first chunk, while the script reported success.
Volume guard nullified single-case sites	Low-volume facilities scored null rather than low, vanishing from the ranking.
A validation that validated nothing	The DRG filter probe accepted a query that matched zero rows, which is precisely how the wrong codes survived.

Four further defects were fatal rather than silent: round() applied to a double-precision value (no such function in PostgreSQL, so the scoring table could never be built), an unguarded length-of-stay cast, a categorical field typed as an integer, and the assumption that a CMS CSV header sits on row one when the template puts it on row three.

4. What the Live Data Actually Looks Like

Every figure below was measured, not estimated.

4.1 Clinical cohort

Measure	2022	2024
Statewide elective hip + knee replacements	30,067	26,980
NYC five-borough discharges	16,679	14,661
NYC facilities performing the procedure	45	47
Facilities clearing the volume floor of 10	40	41
Coded type_of_admission = Elective	93.5%	85.3%

Severity distribution is heavily weighted to the least complex tier — roughly 19,700 of 30,067 statewide discharges at severity 1 in 2022, and only 99 at severity 4 — which is exactly what an elective cohort should look like, and confirms the scope is behaving as intended.

4.2 The market is concentrated

Top NYC facilities by elective joint replacement volume (2022):

Facility	Discharges
Hospital for Special Surgery	6,155
NYU Langone Orthopedic Hospital	1,930
Montefiore Medical Center — Wakefield	1,720
NewYork-Presbyterian / Queens	937
Mount Sinai West	642
Lenox Hill Hospital	614

Hospital for Special Surgery alone accounts for roughly 37 percent of the five-borough market. Any analysis that lost HSS to a name-matching failure would be describing a different city.

4.3 Price transparency compliance

Of 18 target health systems, 14 resolved through the `cms-hpt.txt` protocol, yielding more than 70 individual machine-readable file locations. Multi-campus systems dominate that count: Northwell publishes 24 locations, Hospital for Special Surgery 13, NewYork-Presbyterian and Mount Sinai 9 each.

The four failures are all external, and all instructive about real-world compliance:

System	Reason
Montefiore	Expired TLS certificate
SUNY Downstate	Certificate hostname mismatch
NYC Health + Hospitals	IP-based bot blocking (403)
Wyckoff Heights	No <code>cms-hpt.txt</code> deployed (404)

Confirmed price extractions include Hospital for Special Surgery (90 DRG-470 lines; discounted cash \$79,243.50, negotiated rates \$69,635–\$76,712) and Maimonides Medical Center (60 lines; negotiated \$46,413.19). That spread — a 1.7x difference between two hospitals in the same city for the same procedure — is the market inefficiency this project exists to expose.

5. Architecture and Tool Stack

Every component is free at the tier this project requires. No credit card is needed at any point.

Layer	Service	Cost	Role
ETL	Python 3.14	Free	Socrata API ingestion; streaming MRF parser
Database	PostgreSQL 18 (local)	Free	Staging, transformation, algorithm execution
Repository	GitHub	Free	Public code, and the deployment trigger
Web application	Streamlit Community Cloud	Free	The live consumer-facing URL
Geographic map	Tableau Public	Free	ZCTA choropleth with shapefile joins
Cloud database	Neon (optional)	Free tier	A live-data story instead of a snapshot
Product design	Figma	Free	Consumer UI wireframes

Two deliberate architectural choices

Snapshot before service. The scored output is a few dozen rows. The web app ships that as a committed CSV and treats a cloud database as an optional upgrade, so the public demo cannot break because a free-tier database was suspended for inactivity. Standing up a database server to serve fifty static rows is architecture theatre; the fallback path is the honest answer, and the code demonstrates both.

Two surfaces, not one. Streamlit handles ranking, filtering and cost-versus-quality comparison. Tableau handles geography, because it joins shapefiles natively and Streamlit does not. Neither tool is asked to do the thing it is bad at.

6. Data Integration Methodology

6.1 Clinical outcomes — NY SPARCS

New York mandates that every Article 28 facility report patient-level detail on discharge. The de-identified Public Use File is retrieved through the Socrata Open Data API on Health Data NY.

- Filtered server-side to APR-DRG 324 and 326, so the multi-million-row statewide file is never materialised locally.
- Paged with a stable identifier ordering. A single request with a fixed limit silently truncates the cohort with no way to distinguish a complete pull from a clipped one.
- Two release years are stacked by default, raising per-facility volume and stabilising the ratios for smaller hospitals.

- The Permanent Facility Identifier and operating certificate number are both extracted, because entity resolution depends on them.

De-identification constrains what is possible, and the design accounts for it rather than working around it: dates reduced to a discharge year, ZIP codes truncated to three digits and blanked for small-population prefixes, ages banded into categories, and any stay of 120 days or more collapsed to the literal string 120+.

6.2 Financial data — CMS Hospital Price Transparency

The 2024 OPPS Final Rule made programmatic pricing collection feasible by mandating both a standard file schema and a discovery mechanism: a `cms-hpt.txt` file at the root of each hospital domain pointing a crawler at the machine-readable file.

- The discovery parser reads *every* repeating block, not just the first. Multi-campus systems publish one block per operating certificate, so reading only the first discards most of the market — 1 location instead of 24, in Northwell's case.
- All three permitted formats are parsed — CSV tall, CSV wide, and JSON — and every one streams.
- Format is decided by sniffing the first byte, not the file extension. Maimonides serves JSON from a `download.aspx` endpoint as `application/octet-stream`.
- The CMS CSV template places file metadata on the first two rows and real column headers on the third.
- A 403 triggers one retry with a browser user-agent; hosts that reject both need a manual URL.
- A courtesy delay separates requests. The protocol invites automated collection; that is not licence to hammer a hospital's web server.

6.3 Entity resolution — the crosswalk

State and federal filings rarely spell a hospital identically, so a direct string join loses most of the market. Resolution runs in three tiers: exact match on a normalised name, fuzzy match above a similarity threshold, and manual override. Pairings below the auto-accept threshold are written with a review flag and exported for human confirmation; once confirmed, they are never overwritten by a later automated rebuild.

Uniqueness is enforced at the database level. This is not defensive decoration — two facilities mapped to one CMS name silently fan out the master join and corrupt the market median that the entire financial term is measured against.

7. The Risk-Adjusted Algorithm

7.1 Indirect standardization

Ranking hospitals on raw averages punishes the ones that accept difficult cases. A regional referral centre treating medically fragile patients will always show longer stays than a boutique clinic operating exclusively on healthy candidates, and a marketplace reporting that as a quality difference is actively misleading.

Each discharge is therefore compared against the average for its own APR severity tier, from 1 (Minor) to 4 (Extreme). Averaging those tier expectations across a facility produces its case-mix-weighted expected rate. Observed divided by expected gives the O/E ratio, where below 1.0 means better than expected.

Benchmarks are stratified by severity tier *and discharge year* together. Average length of stay for joint replacement has fallen steadily as same-day discharge protocols spread, so a benchmark pooled across release years would flatter every facility in the later year and penalise every facility in the earlier one — a calendar artefact presented as a quality finding.

7.2 The composite clinical term

Clinical performance is the geometric mean of two O/E ratios: length of stay, and adverse discharge disposition. Geometric rather than arithmetic because both inputs are ratios centred on 1.0, so the mean must be scale-free — twice the expected length of stay against half the expected adverse rate should cancel to 1.0, which only the geometric mean delivers.

Adverse disposition is defined by an explicit allow-list: only discharge to self-care, or to home with home-health support, counts as a good outcome. This matters more than it sounds. The intuitive test of whether the disposition text contains the word "Home" classifies both `Skilled Nursing Home` and `Hospice - Home` as clinical successes.

7.3 The Value Index

```
Value Index = (1 / clinical O/E)
              x (market median cost / facility cost)
              x ln(1 + volume)
```

- **Clinical multiplier** — the inverse of the risk-adjusted O/E. A facility performing at 0.8 of expectation scores 1.25; one at 1.5 scores 0.67.

- **Financial multiplier** — the market median divided by the facility price. The market median is taken across facility-level medians, one vote per hospital, so a system publishing forty payer contracts cannot outvote one publishing three.
- **Experience modifier** — the natural log of one plus procedure volume. Logarithmic so established centres are rewarded without overwhelming cost and quality. The offset is not cosmetic: plain $\ln(\text{volume})$ evaluates to zero at a volume of one and annihilates the entire product.

7.4 Worked example

Facility	Volume	O/E	Cost	Market median	Value Index
High-volume, efficient	420	0.87	\$52,000	\$68,000	9.08
Low-volume, expensive	95	1.12	\$91,000	\$68,000	3.04

The spread is intentional. A facility that is cheaper, faster than its case mix predicts, and more experienced should separate clearly from one that is none of those things.

8. Deployment Plan

Sequenced so each step is verifiable before the next depends on it.

#	Step	Gate
1	Supply the PostgreSQL password	<code>.env</code> populated; connection succeeds
2	Run the SQL smoke test	Prints ALL SMOKE TESTS PASSED
3	Execute the full pipeline	QA diagnostics reviewed; fan-out guard returns nothing
4	Review the facility crosswalk	Fuzzy matches confirmed or corrected by hand
5	Run the web app locally	Renders correctly against real data
6	Commit and push	Credential scan clean
7	Deploy to Streamlit Community Cloud	Live URL loads
8	Publish the Tableau dashboard	Map renders with the ZCTA join
9	Link both from the README	A reader reaches the live product in one click

Two cautions worth stating in advance. Anything saved to Tableau Public is publicly visible by design — acceptable here because every input is already public, but never point it at anything private. And if a credential is ever committed by accident, rotate it immediately; rewriting git history does not un-publish what has already been pushed.

9. Revised Roadmap

Week	Objective	Deliverable
Week 1	Database and first full run	Smoke test passing; SPARCS and MRF data loaded; QA reviewed
Week 2	Data quality and entity resolution	Crosswalk reviewed; scored table trusted
Week 3	Publication	Streamlit app live; snapshot committed; README linked
Week 4	Geographic visualisation	Tableau Public dashboard with shapefile join and tooltips
Week 5	Product design	Figma wireframes translating the index into a five-star interface
Week 6	Portfolio assembly	Screenshots, written methodology, resume entry

10. Known Limitations

Stated plainly, because a portfolio project that overclaims is easier to dismantle in an interview than one that names its own boundaries.

- **Length of stay is a proxy, not an outcome.** The SPARCS public file carries no readmission, revision or complication data. Length of stay and discharge disposition are the best available stand-ins, and both reflect discharge-planning culture as much as surgical quality.
- **The index has never been calibrated.** No one has checked whether a facility scoring 8.0 produces better results than one scoring 4.0. The three terms are multiplied unweighted because that is a defensible starting point, not because evidence says those are the right weights.
- **Long stays are censored.** Anything at or beyond 120 days is recorded identically, so observed length of stay is a floor.
- **Geography is coarse.** Three-digit ZIP prefixes indicate patient catchment, not facility location, and are blanked entirely for small populations.
- **Compliance is partial and non-random.** The market median is computed over the hospitals that chose to comply, which is not a random sample. NYU Langone publishes APR-DRG without 324 or 326 and may show no price at all; Mount Sinai's CDN rejects crawlers and needs a manually sourced URL.
- **Posted rates are not out-of-pocket cost.** Deductibles, coinsurance and benefit design are absent, and the figures cover the facility charge only — not the surgeon, anaesthesiology, separately billed implants, or post-acute care.
- **Stacked years are not uniformly observed.** A facility appearing in only one release has its volume counted over a shorter window, understating its experience modifier.

11. Glossary

Term	Meaning
APR-DRG	All Patient Refined Diagnosis Related Group. The 3M classification used by New York SPARCS. Carries a severity-of-illness subclass from 1 to 4, which is what makes risk adjustment possible. 324 = elective hip replacement, 326 = elective knee replacement.
MS-DRG	Medicare Severity DRG. The federal classification used for reimbursement and in most hospital price files. 470 = major hip or knee joint replacement without major complications. <i>A different system from APR-DRG: the same number means a different procedure.</i>
PFI	Permanent Facility Identifier. New York's stable numeric identifier for a licensed facility. The join key, because facility names change spelling between releases.
O/E ratio	Observed divided by Expected. Below 1.0 means a facility performed better than its patient mix predicted. The standard construction in risk-adjusted quality measurement.
Indirect standardization	Deriving an expected rate by applying population-wide averages to a facility's own case mix, rather than comparing raw averages directly.
Severity of illness	An APR-DRG subclass from 1 (Minor) to 4 (Extreme) describing physiological decompensation. The stratification variable for every benchmark here.
SPARCS	Statewide Planning and Research Cooperative System. New York's mandatory all-payer discharge database, published de-identified.
MRF	Machine-Readable File. The standard-charges file every US hospital must publish under 45 CFR Part 180.
cms-hpt.txt	A file at a hospital's domain root pointing automated crawlers at its MRF. Introduced by the 2024 OPPS Final Rule.
Censoring	Recording all values beyond a threshold identically. SPARCS collapses every stay of 120+ days to one bucket, so observed averages are floors.
Adverse disposition	Any discharge destination other than self-care or home with home health — skilled nursing, inpatient rehab, acute transfer, hospice, or death.
Fan-out	A join that multiplies rows because the key is not unique on one side. The failure mode most likely to corrupt an aggregate silently.

12. Success Criteria

The project is complete when a reader who has never seen it can, in under two minutes:

- Open a live URL and compare NYC hospitals on cost and risk-adjusted quality for a knee replacement.
- See the geographic distribution of value across the five boroughs.
- Read the repository and understand how public clinical and pricing data became that ranking.
- Find a clear statement of what the data cannot support.

The last of those matters as much as the first three. Demonstrating that you know where a model stops being trustworthy is the difference between an analyst and someone who produces charts.

This project is a portfolio demonstration. It is not medical or financial advice and has not been clinically validated. See `DISCLAIMER.md` in the repository.